Artificial Intelligence Component in the Tomato Plant Health Assessment Project

1. General Overview of the AI Component

This project aims to develop an intelligent system based on Machine Learning techniques to analyze agricultural sensor data and predict the health status of tomato plants.
The importance of this system lies in the growing need for data-driven solutions in agriculture, rather than relying solely on human experience or visual inspection, which may be delayed or inaccurate.

The AI component acts as a **decision-making engine** within the application, transforming raw numerical sensor readings into a clear and understandable outcome for farmers: whether the plant is healthy or requires intervention.

2. Problem Definition

The problem addressed by this project can be summarized as follows:

- Agricultural sensor readings (moisture, salinity, minerals, etc.) are raw numerical values that are difficult to interpret directly
- Farmers require fast and clear decisions, not just data
- There is no direct mapping between sensor values and plant health status
- The solution must be practical and easily integrated into a real-world application

Based on this, the objective was to build an AI model capable of **learning the relationship between soil properties and plant health**.

3. Dataset Collection

The first step in developing the AI component was searching for a suitable dataset that aligns with the project's objectives.
The search focused specifically on datasets related to tomato plants, as the entire project revolves around assessing the health of this crop.

The team faced notable challenges in:

- Finding open-source datasets
- Ensuring the dataset contained relevant agricultural features

- Ensuring the data reflected realistic plant conditions

After several attempts, a tomato-related dataset was successfully obtained, containing features suitable for analyzing plant health.

4. Data Quality Assessment

Once the dataset was acquired, an initial quality assessment was performed.
The analysis showed that:

- The data was well-structured
- No missing values were present
- No obvious outliers were detected
- All features were numerical and ready for use

As a result, no additional preprocessing steps (such as data cleaning or imputation) were required, and the workflow proceeded directly to modeling.

5. The Core Challenge: Data Labeling

Although the dataset was clean and usable, the main challenge was not technical implementation but **data labeling**.

The dataset did not include labels indicating:

- Healthy plants
- Unhealthy plants

This meant that:

- Supervised learning could not be applied directly
- No ground truth was available
- Proper labeling required agricultural domain expertise

This stage represented the most critical challenge of the entire project.

6. Domain Expert Consultation

To address the labeling problem, the team consulted experts from the **Faculty of Agriculture at the university**.
A meeting was held with the Dean of the faculty, during which the following were discussed:

- The project idea
- The nature of the available data
- The goal of the AI component

The experts clarified that:

- Accurate plant health classification requires long-term field studies
- A definitive judgment cannot be made solely from sensor values

However, they provided:

- General guidelines
- Preliminary indicators
- Approximate criteria that could be used for initial classification

This step was essential in grounding the technical solution in real agricultural knowledge.

7. Adopted Solution: Using Clustering to Generate Labels

Due to the absence of labeled data, **Unsupervised Learning** techniques were adopted, specifically **Clustering algorithms**.

The solution strategy was as follows:

- Apply clustering to group similar data points
- Divide the dataset into two clusters
- Assume that each cluster represents a different plant health condition

This approach relies on the assumption that:

Similar agricultural feature values typically correspond to similar plant health states.

8. Interpreting Clusters and Mapping Them to Health Status

After applying clustering:

- Each cluster was analyzed independently
- Feature distributions were compared against the criteria provided by agricultural experts

Based on this analysis:

- One cluster was labeled as **Healthy Plants**
- The other cluster was labeled as **Unhealthy Plants**

Through this process, a preliminarily labeled dataset was created, enabling the use of supervised learning techniques.

9. Supervised Machine Learning Model Training

With labels available, multiple supervised learning models were trained to compare performance and select the most suitable one for the dataset.

The models included:

- Support Vector Machine (SVM)
- Random Forest
- Other algorithms used for benchmarking

All models were trained and evaluated using the same dataset to ensure a fair comparison.

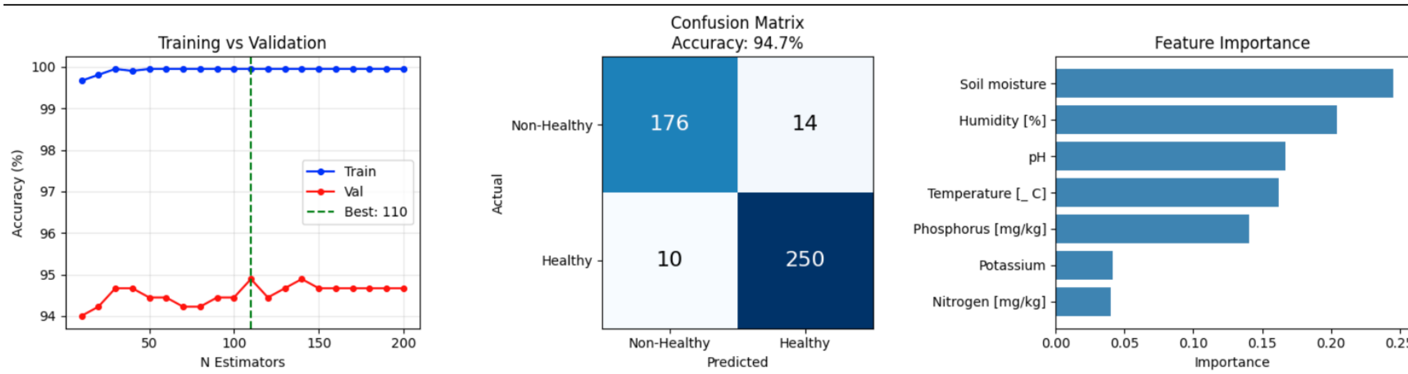
10. Model Evaluation and Results

Model performance was evaluated using a test dataset and standard classification metrics.

The results showed that:

The Random Forest model achieved the best overall performance compared to the other models,
in terms of prediction accuracy, robustness, and ability to handle data complexity.

Evaluation Results:



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Best n_estimators: 110 (Val Accuracy: 94.89%)

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FINAL RESULTS
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Test Accuracy: 94.67%

Classification Report:
              precision    recall  f1-score   support

 Non-Healthy       0.95      0.93      0.94       190
     Healthy       0.95      0.96      0.95       260

    accuracy                           0.95       450
   macro avg       0.95      0.94      0.95       450
weighted avg       0.95      0.95      0.95       450
```

11. System Integration

The trained model was designed to be easily integrated into the agricultural application. The workflow is as follows:

- Sensor readings are received by the system
- The AI model processes the input data
- A health prediction is generated
- The result is displayed to the farmer in a simple and intuitive manner

12. Added Value of the AI Component

The AI component adds significant value to the project by:

- Converting raw sensor data into actionable decisions
- Supporting early agricultural decision-making
- Reducing potential crop losses
- Promoting smart agriculture practices

13. Future Enhancements

Potential future improvements include:

- Training on real-world data collected from farms
- Improving labeling accuracy through extended expert collaboration

- Integrating plant image analysis (Computer Vision)
- Providing actionable recommendations instead of diagnosis only